

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented new features for a core operating system update client, enhancing discovery and installation reliability for **Windows software updates**.
- Collaborated with cross-functional teams across WSD and Windows to build features for efficient component updates, integrating **ML/AI models** and applications.
- Developed and tested features based on internal and external customer requirements within a fast-paced quarterly planning cycle, contributing to organizational impact through process improvements.
- Addressed security vulnerabilities by architecting resilient components for a highly secure and evolving landscape, ensuring the integrity of the update platform.
- Monitored monthly retail releases to Windows, diagnosing and mitigating issues expeditiously to maintain service stability for millions of users.
- Mentored junior developers and interns on the team, providing guidance on coding best practices and project execution.
- Containerized update client services using **Docker** and deployed to an **AWS EKS** cluster, managing configurations and resource allocation for predictable rollouts.

SKILLS

Languages – Java, Python, C++, C#, JavaScript, R

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes (EKS), Jenkins, Git

Backend & Scripting – Spring Boot, REST APIs, Flask, FastAPI, Python scripting, ETL, data pipelines

Databases – PostgreSQL, MySQL, Oracle, SQL

Analytics & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, R/RStudio

Operating Systems – Linux, Shell scripting

Methodologies – Agile, Scrum, CI/CD, Incident Response, Change Management

PROJECTS

Windows Update Client Reliability Enhancement

- Developed a Java-based client module for the Windows update platform, focusing on improving the reliability of update discovery and installation processes across diverse system configurations.
- Implemented enhanced error handling and retry mechanisms within the update lifecycle management code, reducing critical failures by an estimated **18%** during internal testing phases.
- Integrated with **AWS CloudWatch** for detailed logging and monitoring of update client performance, providing actionable insights for continuous improvement and issue resolution.

Automated Data Pipeline for Feature Integration

- Engineered a Python data pipeline to automate the ingestion and preprocessing of data for ML model integration into the update platform, handling approximately **5,000** records per hour.
- Designed and implemented REST APIs using **FastAPI** to serve preprocessed features to downstream ML inference services, ensuring efficient data access and compatibility.
- Wrote comprehensive unit tests using **pytest** for the pipeline components, achieving over **90%** code coverage and ensuring data integrity throughout the ETL process.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)